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T.E - (IT) (Sem-VI)(Revised Course 2019-2020)

EXAMINATION May 2024

IT610 Principles of Compilers

[Time: 3:00 Hours]

[Max. Marks:100]

Instructions: Answer any five questions by selecting two questions each from Part A and Part B, and one question from Part C

PART A

- Q1** a) Compile the following set of C statements by showing the input and output of each phase of compilation 10
float a=10;
for (int i=0; i<10; i++)
{ a=a*i; }
- b) Consider the lex program: 07
% {A, B%}
%%
{ab} {yylval = 1; return(A);}
{a+b*} {yylval = 1; return(A);}
{a* b +} {yylval = 1; return(A);}
%%
Give the implementation of Lexical Analyzer using DFA
- c) Differentiate between Compilers and Interpreters 03
- Q2** a) Check if the following grammar is LL (1) grammar 08
 $S \rightarrow iEtS|iEtSeS|a$
 $E \rightarrow b$
Also define the FIRST and FOLLOW procedure using appropriate examples.
- b) Construct a canonical parsing table for the following grammar: 08
 $S \rightarrow E\#$
 $E \rightarrow E + T|T$
 $T \rightarrow i | (E)$
Also draw a go to graph for the items obtained.
- c) Explain the need for left factoring a grammar with the help of an example. 04
- Q3** a) Explain Bootstrapping with appropriate T diagrams. 04
b) Consider the grammar; 08

$S \rightarrow E = E$

$E \rightarrow E + T \mid T$

$T \rightarrow T * id \mid id$

- i. Construct LR (1) items of the given grammar
- ii. Construct a parsing table and check if the string $id=id+*id$ can be parsed Or not.
- c) List and explain various tools used for Compiler Constructions.
- d) i) Eliminate Left Recursion:

$A \rightarrow Ac \mid Aad \mid bd \mid c$

- ii) Define a handle

04

04

PART B

- Q4 a) Convert the following code to quadruples and triples:
- $y = a * a + 2 * a * b + b * b$
- 08
- b) Explain the different types of errors encountered during the compilation process. 06
- c) Describe the structure of an activation record. 06
- Q5 a) Write the three-address code for the code snippet given below. Construct the basic blocks and draw the flow graph. Further, optimize the code. 10
- ```

sum=0;
for (int j=0; j<=100; j++) {
 sum=sum+1;
 t=50;
 a[j]=sum;
}

```
- b) What is the significance of DAG in compiler construction? Construct a DAG for the following expression. 10
- $count = X[i] * Y + Z[i] * a/b + c$
- Q6 a) Explain the different storage allocation strategies 05
- b) Explain the different code optimization techniques with examples 06
- c) Write a Short note on: 09
- 1) Symbol Table
  - 2) Peephole optimization
  - 3) Next Use Information



**PART C**

- Q7** a) Perform lexical analysis on the following code snippet: **06**  
int b[10];  
int i = 0;  
while(i<5){  
b[i] = 1.0;  
i++; }
- b) Write the translation Scheme for Boolean Statements **06**
- c) Construct a canonical LR parsing table for the below grammar and parse the string  $i + (i + i)$ : **08**  
 $S \rightarrow E \#$   
 $E \rightarrow E + T \mid T$   
 $T \rightarrow i \mid (E)$
- Q8** a) Define input buffering scheme. Explain with the help of a pseudocode the use of sentinel character in the input buffering **06**
- b) Consider the following code statement: **08**  
$$x = ((a+b)-c+(d-e))$$
  - 1) Write the final code generated by the above statement
  - 2) Explain an address descriptor and a register descriptor.
  - 3) Show the contents of the address descriptor and register descriptor at each statement
- c) Write a lex program to validate the following sets of tokens: **06**
  - 1) (letter) (letter | digit) \*
  - 2) (digit)+(digit)+
  - 3) keywords